

LAB 6: SEMAPHORE IMPLEMENTATION

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C PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <semaphore.h>
#include <sys/mman.h>
#include <sys/wait.h>

int main()
{
    sem_t *sem;
    sem = mmap(NULL, sizeof(sem_t), PROT_READ | PROT_WRITE, MAP_SHARED | MAP_ANONYMOUS, -1,
0);
    sem_init(sem, 1, 1);
    if(fork() == 0)
    {
        sem_wait(sem);
        printf("Child Process Entering Critical Section\n");
        sleep(3);
        printf("Child Process Leaving Critical Section\n");
        sem_post(sem);
        exit(0);
    }
    sem_wait(sem);
    printf("Parent Process Entering Critical Section\n");
    sleep(3);
    printf("Parent Process Leaving Critical Section\n");
    sem_post(sem);
    wait(NULL);
    sem_destroy(sem);
    return 0;
}
```

OUTPUT:

```
(kali@Harish)-[~]
$ nano semaphore.sh
(kali@Harish)-[~]
$ chmod +x semaphore.sh
(kali@Harish)-[~]
$ bash semaphore.sh
Entering Critical Section
5
Leaving Critical Section
```